

Subject: Java + Adv Java

Duration: 3Hrs

Total Marks 50

Instructions:

1. All questions are mandatory
 2. Switch off your mobile phones during examination.
 3. You are allowed to refer Java/MSDN Documentation and your assignments.
 4. Any sort of malpractice will be taken seriously and will be reported to the concerned authorities immediately.
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Question 1

ABC India received a project from **Carl Automobile Corporation**. They want to automate their Car Service process to improve customer interaction. As per the initial interaction below given use cases are identified:

1. Submit car repairing request.
2. Display Report.

To implement the above use cases, follow the below given steps.

Step 1 – Design a user interface which has the following menu driven options:

- a) Press 1 to enter the new request
- b) Press 2 to get the request status, by entering requestId
- c) Press 3 to update the car repairing status
- d) Pres 4 Display Admin Report

*[In order to use option no c & d , user must pass the username and password,
username: admin, password: admin4carApp, if invalid throw and Exception "Invalid user"]*

a) On the selection of 1 following option appears which prompts to enter the car repairing request

Please Enter Your Name:

Please Enter Your Car Registration Number:

[The car registration number must be the size of 8 digits and starts from 456XXXXX] , else generate Custom Exception Invalid Registration number, use Regular Expression to compare the Registration number]

Please Enter Car Repairing Date

[For each date, only 5 cars can be registered., otherwise the system should throw a custom exception called SlotNotAvailableException .

The exception message should clearly mention the date entered by the user and say that slots are not available on that date (XX-XX-XXXX). It should also suggest the next available date(s) where slots are still open.]

Please Enter the Problems you are facing in your car:

Press 1 to submit or 2 to exit.

On the submit of above request a record will be entered into the CSV file

“CarServiceRequest.csv” which holds the details of car and its issues.

System also generates a **requestid** corresponding to the request. By default a flag pending will be mentioned and on the completion of service a flag will be changed to Repaired.

CSV file Data should include

Request Id , Date, Car Registration Number, Username, Problem, status

Step 2 – To update the car repairing status system prompt to enter the **requestid** assigned to the car. If the status of the corresponding car is **pending** then it will be changed to **Repaired** else the message appeared “Car corresponding to this requestid is repaired”.

[Implement the following code using Java Stream API]

Step 3 – [Implement all code through java Stream API]

3.1 To display the report in the below mentioned format.

RequestId	RegistrationNo	RepairingDate	Issue	flagstatus
1	UP14-AB-6013	01/12/2013	Overall Service, Break, Tail light, ac check	Pending
2	UP15-AB-2345	03/12/2013	Overall Service, Break, Tail light, Head light check.	Repaired
3	UP16-AB-2345	06/12/2013	Overall Service, Gear, Radio check.	Pending
4	UP18-AB-5678	08/12/2013	Overall Service, ac, Tyres check.	Pending

3.2) filter the car based on Pending Status

3.3) filter the cars based on Date

3.4) filter the cars based on two given Date

3.5) filter the car based on particular requestId

Steps	Marks
Properly designing the user interface & reports	5
Properly submitting the new repairing request in the CSV file	5
All Exception Handling, regular Expression and proper Error Statement (2*3)	6
Updating the car repairing status	5
Displaying report as per the given format.(5 * 5)	25
Proper comments, coding conventions & indentation	4